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Apparatus, method, and medium for mobile access point station

Abstract

Some embodiments include an apparatus and method for a mobile access point (AP) station that operates in an always-on soft AP mode where a station is associated with the mobile AP station, and switch back to operate as a station in an infrastructure mode with an AP. For example, in an always-on soft AP mode, the mobile AP station may implement a target wake time (TWT) responder power save mode feature. The mobile AP station may maintain a wake window after a beacon to enable unassociated devices to associate; establish a broadcast TWT schedule for unassociated TWT-capable devices to associate; schedule TWT for associated devices; go to sleep outside of the scheduled TWT of the associated devices; and switch back to infrastructure mode to perform as a station and maintain a connection with an AP.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. patent application Ser. No. 16/286,156, filed on Feb. 26, 2019, entitled, Apparatus, Method, and Medium for Implementing a Target Wake Time, which claims benefit of U.S. Provisional Application No. 62/692,562, filed on Jun. 29, 2018, entitled, Apparatus and Method for a Mobile Access Point Station, which are both incorporated herein by reference in their entireties.

BACKGROUND

Field

(1) The described embodiments generally relate to channel access in wireless communications.

Related Art

(2) A wireless local area Network (WLAN) access point (AP) transmits infrastructure traffic, is always active, and can be reached whenever associated or unassociated stations choose to communicate with the AP. Being always active, the AP operates in an always-on mode and cannot go into a power-save mode. A mobile device may perform soft AP functions that enable other devices to access a network. Like an AP, a soft AP should be always-on so that associated or unassociated stations can communicate with the soft AP. Thus, a soft AP cannot enter a power save mode. But the mobile device performing soft AP functions is battery-operated, and being always-on causes excessive power consumption on the battery. Consequently, a soft AP function on a mobile device is turned on by a user for a given period and then turned off. Further, a mobile device can

operate as a station in infra-structure mode to transmit infrastructure traffic with an AP. But, once soft AP functions on a mobile device are turned on, the mobile device cannot switch back to infrastructure mode as a station to communicate with the AP; this is because a soft AP is required to be always available to associated stations.

SUMMARY

(3) Some embodiments include an apparatus and method for a mobile access point (AP) station. A mobile AP station is a mobile device that operates in an always-on soft AP mode that provides soft AP mode functions, and is capable of switching back to infrastructure mode to operate as an associated station in an infrastructure mode with an AP. For example, the mobile AP station may implement a target wake time (TWT) responder power save mode feature with stations associated with the mobile AP station. The mobile AP station may: maintain a wake window after each beacon or one or more selected beacons to allow unassociated devices to communicate with the mobile AP station; establish a broadcast TWT schedule for unassociated TWT-capable devices to communicate with the mobile AP station; schedule TWT for associated devices; go to sleep outside of the scheduled TWT of the associated devices; and concurrently perform as a station and maintain a connection (e.g., wireless local area network (WLAN) access) with an AP. Thus, a mobile AP station in an always-on soft AP mode may: go to sleep, switch back to infrastructure mode and communicate infrastructure traffic to the AP, switch back to operate in a soft AP mode and communicate with one of the associated devices of the mobile AP station at a scheduled TWT, or remain active.

(4) Some embodiments include a mobile AP station that may operate in an always-on soft AP mode that includes implementing a TWT responder power save mode. While operating in the always-on soft AP mode, the mobile AP station may enable a first station to associate with the mobile AP station, and while the first station is associated with the mobile AP station, switch back to infrastructure mode to transmit infrastructure traffic to an AP. In addition, the mobile AP station may implement a broadcast TWT schedule, and advertise the broadcast TWT schedule in a beacon. The broadcast TWT schedule indicates a service period during which an unassociated TWT-capable station (e.g., an IEEE 802.11ax device) may wake at a known time window (e.g., when the mobile AP station is available) to associate with the mobile AP station. The mobile AP station may also enable the first station to request setting up a TWT for the first station by setting a TWT Required field to a predetermined value (e.g., “1”) in a high efficiency operation element in a beacon. In response to the TWT Required field being set to the predetermined value (e.g., “1”), the mobile AP station receives a TWT setup request from the first station, and in response to the TWT setup request received, transmits a TWT schedule to the first station. The mobile AP station may inform the first station that the mobile AP station is not available outside of the TWT schedule by setting a TWT Responder Power Save field in a TWT Element, and transmitting the TWT element to the first station. The mobile AP station may enter a sleep mode outside of the TWT for the first station, remain active outside of the TWT for the first station, or dynamically adjust the TWT for the first station. Dynamically adjusting the TWT for the first station may be based in part on a number of stations associated with the mobile AP station, a traffic load between the mobile AP station and one or more stations associated with the mobile AP station, and/or the infrastructure traffic between the mobile AP station and the AP. The mobile AP station may set up a wake window during which new unassociated stations may join a network of the mobile AP station that includes the first station, and where the wake window is temporally adjacent to a beacon. The mobile AP station may dynamically adjust the wake window based at least in part on a number of stations associated with the mobile AP station.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The accompanying drawings, which are incorporated herein and form part of the specification, illustrate the presented disclosure and, together with the description, further serve to explain the principles of the disclosure and enable a person of skill in the relevant art(s) to make and use the disclosure.

(2) FIG. 1 illustrates an example system implementing a mobile access point (AP) station, according to some embodiments of the disclosure.

(3) FIG. 2 illustrates a block diagram of an example wireless system for a mobile AP station, according to some embodiments of the disclosure.

(4) FIG. 3 illustrates an example operation of a mobile AP station with no associated devices, according to some embodiments of the disclosure.

(5) FIG. 4 illustrates an example operation of a mobile AP station with associated devices, according to some embodiments of the disclosure.

(6) FIG. 5 illustrates an example operation of a mobile AP station with unassociated target wake time (TWT)-capable devices, according to some embodiments of the disclosure.

(7) FIG. 6 illustrates an example method for a wireless system supporting a mobile AP station, according to some embodiments of the disclosure.

(8) FIG. 7 is an example computer system for implementing some embodiments or portion(s) thereof.

(9) The presented disclosure is described with reference to the accompanying drawings. In the drawings, generally, like reference numbers indicate identical or functionally similar elements. Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference number first appears.

DETAILED DESCRIPTION

(10) Some embodiments include an apparatus, method, and computer program product for a mobile access point (AP) station that operates in an always-on soft AP mode where a station is associated with the mobile AP station, and operate as a station in an infrastructure mode with an AP. For example, in an always-on soft AP mode, the mobile AP station may implement a target wake time (TWT) responder power save mode feature. The mobile AP station may maintain a wake window after a beacon to enable unassociated devices to associate; establish a broadcast TWT schedule for unassociated TWT-capable devices to associate; schedule TWT for associated devices; go to sleep outside of the scheduled TWT of the associated devices; and perform as a station and maintain a connection with an AP (e.g., to transmit infrastructure traffic to the AP.)

(11) FIG. 1 illustrates an example system **100** implementing a mobile access point (AP) station **150**, according to some embodiments of the disclosure. A mobile AP station **150** is different than a conventional mobile device that performs soft AP functions. Example system **100** is provided for the purpose of illustration only and is not limiting of the disclosed embodiments. System **100** may include but is not limited to stations **120**, access point (AP) **110**, network **130**, and mobile AP station **150**. Stations **120a-120e** may include but are not limited to WLAN stations such as wireless communication devices, smart phones, laptops, desktops, tablets, personal assistants, monitors, and televisions. Stations **120** may support latency sensitive applications (e.g., video and/or audio streaming). AP **110** may include but are not limited to WLAN electronic devices such as a wireless router, a wearable device (e.g., a smart watch), a wireless communication device (e.g., a smart phone), or a combination thereof. Network **130** may be the Internet and/or a WLAN. Infrastructure mode communications are shown as wireless communications **140**. Soft AP mode communications are shown as wireless communications **155**. Mobile AP station **150** may include but is not limited to WLAN stations such as wireless communication devices, smart phones, laptops, tablets, personal assistants, monitors, and televisions.

(12) Mobile AP station **150** operates in an always-on soft AP mode, and is capable of switching

back to infrastructure mode to operate as a station in an infrastructure mode with AP 110. Since mobile AP station 150 operates in an always-on soft AP mode, stations 120b-120e may always discover mobile AP station 150. Stations 120b-120e may associate with mobile AP station 150 and join a network of mobile AP station 150. Concurrent to being in the always-on soft AP mode where a station (e.g., 120b) is associated with mobile AP station 150, mobile AP station 150 can switch back to infrastructure mode to operate as a station in an infrastructure mode, and may transmit and/or receive infrastructure traffic shown as 140b to AP 110. Mobile AP station 150 may use any mechanism such as a Power Management bit in a MAC packet header to inform AP 110 that mobile AP station 150 is going to sleep before switching back to perform soft AP functionality with other devices such as stations 120a-120c. A mobile AP station by default may be in always-on soft AP mode. A user may change a setting, such as manually turning off the always-on soft AP mode.

(13) In some embodiments, when mobile AP station 150 operates in infrastructure mode as a station with AP 110, mobile AP station 150 may arrange a target wake time (TWT) for infrastructure mode with AP 110, or AP 110 may arrange a TWT with mobile AP station 150. Outside of these arranged TWTs with AP 110, mobile AP station 150 may switch back to perform soft AP functionalities with stations 120a-120c that are associated with mobile AP station 150.

(14) FIG. 2 illustrates a block diagram of an example wireless system 200 of a system 100 implementing a mobile AP station, according to some embodiments of the disclosure. System 200 may be any of the electronic devices (e.g., AP 110, station 120, mobile AP station 150) of system 100. System 200 includes central processing unit (CPU) 210, transceiver 220, communication interface 225, communication infrastructure 230, memory 235, and antenna 250. Memory 235 may include random access memory (RAM) and/or cache, and may include control logic (e.g., computer software) and/or data. CPU 210 together with instructions stored in memory 235 performs operations enabling a wireless system 200 of a system 100 implementing a mobile AP station. Transceiver 220 transmits and receives communications signals that support mobile AP station functions according to some embodiments, and may be coupled to antenna 250. Communication interface 225 allows system 200 to communicate with other devices that may be wired and/or wireless. Communication infrastructure 230 may be a bus. Antenna 250 may include one or more antennas that may be the same or different types.

(15) Some embodiments enable a mobile AP station to implement a Target Wake Time (TWT) protocol as a TWT responder, and utilize TWT Responder Power Save Mode features. Using the TWT protocol enables the mobile AP station to remain in the always-on soft AP mode and concurrently maintain a connection with an AP as a station in an infrastructure mode. For example, a mobile AP station may utilize the TWT protocol to: set an individual TWT service period for an associated station to access the medium (e.g., WLAN medium access); and/or arrange for the mobile AP station to go to sleep outside of a TWT service period. When a mobile AP station goes to sleep (e.g., enters a power save mode) transmissions from unassociated stations may be missed. To avoid missing transmissions from unassociated stations, a mobile AP station may stay active for some time after a beacon or selected beacons. The stay active time may be called a wake window for unassociated stations. During a wake window for unassociated stations, the mobile AP station is active and may receive communications from stations such as 802.11 legacy devices and/or TWT-capable devices. The duration of the wake window for unassociated stations can be dynamically adjusted based on a number of stations currently associated with the mobile AP station. For example, when no stations are associated with the mobile AP station, the wake window for unassociated stations can be set to a longer time than when a station is associated. The longer time may increase the chances of unassociated stations being able to join a network of the mobile AP station. Conversely, when many stations are associated with the mobile AP station, the wake window for unassociated stations may be reduced to limit congestion, for example.

(16) Since 802.11 legacy devices do not support TWT capabilities, when an 802.11 legacy device associates with the mobile AP station, the mobile device exits mobile AP station functions.

(17) In addition to a wake window for unassociated stations, a mobile AP station may set up a broadcast TWT schedule and advertise a broadcast TWT service period in a beacon to help TWT-capable devices conserve power usage. For example, TWT-capable devices that receive the beacon can remain asleep until the broadcast TWT service period. The TWT-capable devices may wake during the broadcast TWT to associate with the mobile AP station. Thus, a TWT-capable device may remain asleep and may not have to contend with 802.11 legacy devices during a wake window for unassociated station time period.

(18) FIG. 3 illustrates an example operation **300** of a mobile AP station with no associated devices, according to some embodiments of the disclosure. As a convenience and not a limitation, FIG. 3 may be described with regard to elements of FIG. 1. Operation **300** may represent the operation of mobile AP station **150** when devices **120b-120e** are not associated with mobile AP station **150**, and mobile AP station **150** transmits infrastructure traffic to AP **110**. Operation **300** includes beacons **310** and **340**, wake windows for unassociated stations **315** and **345**, infrastructure modes **320** and **330**, and sleep mode **325** and **335**.

(19) During operation, mobile AP station **150** operates in an always-on soft AP mode and transmits beacon **310**. Subsequent to transmitting beacon **310**, mobile AP station **150** remains active for a duration of wake window for unassociated stations **315** to give 802.11 legacy devices as well as unassociated TWT-capable devices an opportunity to associate with mobile AP station **150**. Mobile AP station **150** may switch to infrastructure mode **320** and transmit and/or receive communications from AP **110**, then go into a power save mode at sleep mode **325**. After sleep mode **325**, mobile AP station **150** may wake to exchange infrastructure traffic again with AP **110** during infrastructure mode **330**, and return to sleep at sleep mode **335**. At the start of the next period, mobile AP station **150** transmits beacon **340** and may dynamically change a duration of wake windows for unassociated stations **345**, based on a number of associated stations **120**. In this example, no stations **120** are associated so mobile AP station **150** may remain awake longer to allow unassociated stations **120** to join a network that includes mobile AP station **150**. Thus, mobile AP station **150** may extend wake window for unassociated stations **345** to be longer than wake window for unassociated stations **315**.

(20) FIG. 4 illustrates an example operation **400** of a mobile AP station with associated devices, according to some embodiments of the disclosure. As a convenience and not a limitation, FIG. 4 may be described with regard to elements of FIGS. 1 and 3. Operation **400** may represent the operation of mobile AP station **150** when devices **120b-120d** are associated with mobile AP station **150**, and mobile AP station **150** is able to switch back to infrastructure mode to transmit infrastructure traffic to AP **110**. Operation **400** includes beacons **410** and **445**, wake windows for unassociated stations **415** and **450**, infrastructure modes **420** and **430**, TWT for TWT-capable associated stations **425** and **435**, and sleep mode **440**. Mobile AP station **150** operates in an always-on soft AP mode and transmits beacon **410**. Mobile AP station **150** remains awake for the duration of wake window for unassociated stations **415**, and switches to infrastructure mode **420** to exchange infrastructure traffic with AP **110**. Subsequently, during TWT for TWT-capable associated stations **425**, mobile AP station **150** may exchange soft AP traffic with TWT-capable associated stations, such as stations **120b-120d**. For example, stations **120b-120d** may contend for access or exchange soft AP traffic as allocated, during TWT for TWT-capable associated stations **425**. Mobile AP station **150** may switch to infrastructure mode **430** and exchange infrastructure traffic with AP **110**, and during TWT for TWT-capable associated stations **435**, exchange soft AP traffic with one or more of associated stations **120b-120d**. Mobile AP station **150** may enter a power save mode during sleep mode **440**. In the next period, mobile AP station **150** may transmit beacon **445** and implement wake window for unassociated stations **450**. Mobile AP station may dynamically adjust the active time after beacon **445** based at least on the number of associated stations **120b-120d**. For example, the time for wake window for unassociated stations **450** may be reduced compared to wake window for unassociated stations **415**.

(21) FIG. 5 illustrates an example operation **500** of a mobile AP station with unassociated TWT-capable devices, according to some embodiments of the disclosure. As a convenience and not a limitation, FIG. 5 may be described with regard to elements of FIGS. 1, 3, and 4. Operation **500** may represent the operation of mobile AP station **150** when devices **120b-120d** are associated with mobile AP station **150**, device **120e** is a TWT-capable device that is not associated with mobile AP station **150**, and mobile AP station **150** is able to switch back to infrastructure mode to transmit infrastructure traffic to AP **110**. Operation **500** includes beacons **510** and **550**, wake windows for unassociated stations **515** and **555**, infrastructure mode **520**, and sleep modes **530** and **549**. Mobile AP station **150** may group associated stations **120b-120d** to different TWT periods to reduce contention in a TWT period. In operation **500**, stations **120b** and **120c** may be assigned to TWT for associated stations **425** and station **120d** may be assigned to TWT for associated station **535**. Mobile AP station **150** may arrange a broadcast TWT schedule that is advertised in beacon **510**. Unassociated TWT-capable stations like station **120e** may receive the broadcast TWT schedule and remain asleep until a broadcast TWT service period such as TWT for unassociated TWT-capable stations **545**.

(22) In operation **500**, Mobile AP station **150** operates in an always-on soft AP mode and transmits beacon **510**. Mobile AP station **150** remains awake for the duration of wake window for unassociated stations **515**, and switches to infrastructure mode **520** to exchange infrastructure traffic with AP **110**. Subsequently, during TWT for TWT-capable associated stations **525**, mobile AP station **150** may exchange soft AP traffic with TWT-capable associated stations **120b** and **120c**. For example, stations **120b** and **120c** may contend for access or exchange soft AP traffic as allocated, during TWT for TWT-capable associated stations **525**. Mobile AP station **150** may enter and remain in a power save mode for sleep mode **530**, and then exchange soft AP traffic during TWT for TWT-capable associated stations **535** with associated station **120d**. Mobile AP station **150** may enter and remain in a power save mode during sleep mode **540**. At TWT for unassociated TWT-capable stations **545**, mobile AP station **150** may remain active to enable unassociated TWT-capable stations like station **120e** to join the network of mobile AP station **150**.

(23) Based on the activity in the period such as station **120e** associating with mobile AP station **150**, mobile AP station **150** may adjust the time period of wake window for unassociated stations **555** and/or a duration of the next TWT for unassociated TWT-capable stations time period. Mobile AP station **150** may also assign station **120e** to TWT for station **525** or **535**. Mobile AP station **150** may also adjust the TWT schedule as well as a duration of the infrastructure mode based on any/all of: a number of stations associated with mobile AP station **150**, a soft AP traffic load of the associated stations **120** associated with mobile AP station **150**, and/or an infrastructure traffic load of mobile AP station **150** and AP **110**.

(24) FIG. 6 illustrates an example method **600** for a wireless system supporting a mobile AP station, according to some embodiments of the disclosure. As a convenience and not a limitation, FIG. 6 may be described with regard to elements of FIGS. 1-5. Method **600** may represent the operation of mobile AP station **150** of FIG. 1 performing always-on soft AP functions and also being able to switch back to infrastructure mode to transmit infrastructure traffic to AP **110** as a station. Method **600** may also be performed by system **200** of FIG. 2 or computer system **1000** of FIG. 10.

(25) At **605**, system **200** implements a TWT protocol as a TWT responder, and utilizes TWT Responder Power Save Mode features. Using the TWT protocol enables system **200** to remain in the always-on soft AP mode and concurrently maintain a connection with an AP as a station in an infrastructure mode. For example, mobile AP station **150** may utilize the TWT protocol to: set an individual TWT service period for one or more associated TWT-capable stations **120** to access the WLAN medium via communications **155a-155c**, set a time for transmitting infrastructure data to AP **110** as a station via communications **140b**, set a time for unassociated devices to communicate with mobile AP station **150**, set a time to sleep, and/or set a time to remain awake. Mobile AP

station **150** may set up individual TWT with each associated station by: setting and transmitting a specific field in a beacon that indicates that stations are to initiate a TWT setup request with the mobile AP station **150**; and/or if the mobile AP station **150** has not received a TWT setup request from an associated station **120**, the mobile AP station may set up an unsolicited TWT with a station **120**.

(26) In an embodiment, system **200** transmits a high efficiency (HE) Operation Element in a beacon, that instructs receiving stations to request setting up a TWT with mobile AP station **150**. In particular, the HE Operation Element includes a TWT Required Field set to “1.” As an example, CPU **210** together with instructions stored in memory **235** may set the TWT Required Field of HE Operation Element to “1” enabling system **200** (e.g., mobile AP station **150**) to transmit a beacon that includes the HE Operation Element with a TWT Required field set to “1” using the transceiver **220**.

(27) At **610**, system **200** sets up a wake window for unassociated station, including a number and duration that occur after one or more select beacons. Examples of a wake window for unassociated station include **315** and **345** of FIGS. **3**, **415** and **450** of FIG. **4**, and **515** and **555** of FIG. **5**. The number and duration of wake window for unassociated stations may be dynamically updated based at least on a number of stations that associate with system **200** (e.g., mobile AP station **150**.) For example, when there are no stations or few stations associated with mobile AP station **150**, then system **200** may increase a number of and/or duration of wake windows for unassociated stations to allow new unassociated stations **120** to join a network of mobile AP station **150**. When there are many stations associated with mobile AP station **150**, then system **200** may decrease a number of and/or duration of wake windows for unassociated stations to control the number of new unassociated stations **120** that may join the network of mobile AP station **150**. A wake window for unassociated station may be scheduled adjacent to a beacon or specific beacons (e.g., **310** and **340**, **410** and **445**, and/or **510** and **550**.)

(28) At **620**, system **200** may set up a broadcast TWT schedule advertised in a beacon, where the TWT schedule includes a number and duration of service periods for TWT for unassociated TWT-capable stations. TWT-capable devices that receive the beacon can remain asleep, avoid contention with devices during a wake window for unassociated station **515** of FIG. **5**, and wake to associate with mobile AP station **150** at TWT for unassociated TWT-capable stations **545** of FIG. **5**.

(29) At **625**, system **200** determines a schedule for a number and duration of infrastructure mode periods. Concurrent with maintaining an association with a station (e.g., station **120b**) in an always-on soft AP mode, mobile AP station **150** can switch back to infrastructure mode as a station and exchange infrastructure traffic with AP **110**. Based on the load and the time critical aspects infrastructure traffic, mobile AP station **150** determines a number and duration of periods for operating as a station to communicate with AP **110**. Examples of infrastructure modes include **320** and **330** of FIGS. **3**, **420** and **430** of FIG. **4**, and **520** of FIG. **5**.

(30) At **630**, system **200** operates according to the set ups, TWT schedule, and infrastructure mode duration described above. Examples of the operations are described in FIGS. **3-5**.

(31) At **635**, system **200** associates with one or more stations **120**.

(32) At **645**, system **200** schedules a TWT or an unsolicited TWT for a TWT-capable station. For example, in response to receiving the beacon, station **120c** may initiate a TWT setup request and system **200** responds with an assignment of a TWT service period. Example assignments include **425** and **435** of FIGS. **4** and **525** and **535** of FIG. **5**. If system **200** does not receive a TWT setup request from the TWT-capable station (e.g., station **120d**) system **200** may schedule an unsolicited TWT for station **120d**, e.g., **535** of FIG. **5**.

(33) At **650**, system **200** informs associated stations that mobile AP station **150** is not available outside of the scheduled TWT. Mobile AP station **150** may arrange to go to sleep outside of a scheduled TWT service period (e.g., sleep modes **530** and **540** of FIG. **5**.) There may be less stations associated with mobile AP station **150** than AP **110**, thus after TWT schedules are arranged,

mobile AP station **150** may have time to go to sleep outside of the scheduled TWT service periods. To arrange to go to sleep outside of a TWT service period, mobile AP station **150** may set TWT Responder Power Save bit during TWT setup. Implementing TWT responder power save mode features enable mobile AP station **150** to reduce power consumption. As an example, system **200** may transmit to an associated station **120** a TWT Element that includes a TWT Responder Power Save field set to “1.” CPU **210** together with instructions stored in memory **235** (e.g., a driver) may set the TWT Responder Power Save field of TWT Element to “1” enabling system **200** (e.g., mobile AP station **150**) to transmit to an associated station, a TWT Element that includes a TWT Responder Power Save field set to “1.” Thus, station **120** will be aware that mobile AP station **150** may go to sleep outside of a TWT service period. Consequently, associated TWT-capable stations **120** may not communicate with mobile AP station **150** outside of their TWT service period. Using the HE Operation Element with the TWT Required Field set to “1” and the TWT Element that includes a TWT Responder Power Save field set to “1,” system **200** can set up a TWT with each associated station and may go to sleep outside of a scheduled TWT service period.

(34) At **655**, system **200** dynamically adjusts TWT schedule, infrastructure mode duration, broadcast TWT schedule, and/or wake window for unassociated stations. For example, the TWT parameters and durations of time may be adjusted based on a number of stations associated with mobile AP station **150**, a traffic load between associated stations **120b-120d** and mobile AP station **150**, and/or the infrastructure traffic load between the mobile AP station **150** and AP **110**. In addition, system **200** dynamically adjusts a number and duration of service periods for wake windows for unassociated stations based at least on the number of stations **120** associated with system **200** (e.g., mobile AP station.) Method **600** returns to **630** and operates accordingly.

(35) Various embodiments can be implemented, for example, using one or more computer systems, such as computer system **700** shown in FIG. 7. Computer system **700** can be any well-known computer capable of performing the functions described herein such as devices **110**, **120**, or **150** of FIG. 1. Computer system **700** includes one or more processors (also called central processing units, or CPUs), such as a processor **704**. Processor **704** is connected to a communication infrastructure **706** (e.g., a bus.) Computer system **700** also includes user input/output device(s) **703**, such as monitors, keyboards, pointing devices, etc., that communicate with communication infrastructure **706** through user input/output interface(s) **702**. Computer system **700** also includes a main or primary memory **708**, such as random access memory (RAM). Main memory **708** may include one or more levels of cache. Main memory **708** has stored therein control logic (e.g., computer software) and/or data.

(36) Computer system **700** may also include one or more secondary storage devices or memory **710**. Secondary memory **710** may include, for example, a hard disk drive **712** and/or a removable storage device or drive **714**. Removable storage drive **714** may be a floppy disk drive, a magnetic tape drive, a compact disk drive, an optical storage device, tape backup device, and/or any other storage device/drive.

(37) Removable storage drive **714** may interact with a removable storage unit **718**. Removable storage unit **718** includes a computer usable or readable storage device having stored thereon computer software (control logic) and/or data. Removable storage unit **718** may be a floppy disk, magnetic tape, compact disk, DVD, optical storage disk, and/or any other computer data storage device. Removable storage drive **714** reads from and/or writes to removable storage unit **718** in a well-known manner.

(38) According to some embodiments, secondary memory **710** may include other means, instrumentalities or other approaches for allowing computer programs and/or other instructions and/or data to be accessed by computer system **700**. Such means, instrumentalities or other approaches may include, for example, a removable storage unit **722** and an interface **720**. Examples of the removable storage unit **722** and the interface **720** may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as

an EPROM or PROM) and associated socket, a memory stick and USB port, a memory card and associated memory card slot, and/or any other removable storage unit and associated interface.

(39) Computer system **700** may further include a communication or network interface **724**.

Communication interface **724** enables computer system **700** to communicate and interact with any combination of remote devices, remote networks, remote entities, etc. (individually and collectively referenced by reference number **728**). For example, communication interface **724** may allow computer system **700** to communicate with remote devices **728** over communications path **726**, which may be wired and/or wireless, and which may include any combination of LANs, WANs, the Internet, etc. Control logic and/or data may be transmitted to and from computer system **700** via communication path **726**.

(40) The operations in the preceding embodiments can be implemented in a wide variety of configurations and architectures. Therefore, some or all of the operations in the preceding embodiments may be performed in hardware, in software or both. In some embodiments, a tangible, non-transitory apparatus or article of manufacture includes a tangible, non-transitory computer useable or readable medium having control logic (software) stored thereon is also referred to herein as a computer program product or program storage device. This includes, but is not limited to, computer system **700**, main memory **708**, secondary memory **710** and removable storage units **718** and **722**, as well as tangible articles of manufacture embodying any combination of the foregoing. Such control logic, when executed by one or more data processing devices (such as computer system **700**), causes such data processing devices to operate as described herein.

(41) Based on the teachings contained in this disclosure, it will be apparent to persons skilled in the relevant art(s) how to make and use embodiments of the disclosure using data processing devices, computer systems and/or computer architectures other than that shown in FIG. 7. In particular, embodiments may operate with software, hardware, and/or operating system implementations other than those described herein.

(42) It is to be appreciated that the Detailed Description section, and not the Summary and Abstract sections, is intended to be used to interpret the claims. The Summary and Abstract sections may set forth one or more but not all exemplary embodiments of the disclosure as contemplated by the inventor(s), and thus, are not intended to limit the disclosure or the appended claims in any way.

(43) While the disclosure has been described herein with reference to exemplary embodiments for exemplary fields and applications, it should be understood that the disclosure is not limited thereto. Other embodiments and modifications thereto are possible, and are within the scope and spirit of the disclosure. For example, and without limiting the generality of this paragraph, embodiments are not limited to the software, hardware, firmware, and/or entities illustrated in the figures and/or described herein. Further, embodiments (whether or not explicitly described herein) have significant utility to fields and applications beyond the examples described herein.

(44) Embodiments have been described herein with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined as long as the specified functions and relationships (or equivalents thereof) are appropriately performed. In addition, alternative embodiments may perform functional blocks, steps, operations, methods, etc. using orderings different from those described herein.

(45) References herein to “one embodiment,” “an embodiment,” “an example embodiment,” or similar phrases, indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of persons skilled in the relevant art(s) to incorporate such feature, structure, or characteristic into other embodiments

whether or not explicitly mentioned or described herein.

(46) The breadth and scope of the disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A mobile access point (AP) station, comprising: a transceiver configured to transmit and receive wireless communications; and one or more processors communicatively coupled to the transceiver, configured to: operate the mobile AP station in an always-on soft AP mode comprising implementing a target wake time (TWT) feature; transmit a beacon that is followed by a wake window for stations unassociated with the mobile AP station, wherein the beacon comprises a broadcast TWT schedule that indicates a service period during which one or more TWT-capable stations unassociated with the mobile AP station associates with the mobile AP station; while operating the mobile AP station in the always-on soft AP mode, enable a first TWT-capable station to associate with the mobile AP station during the service period, wherein the mobile AP station provides soft AP functionality for the associated first TWT-capable station during a TWT period; while the first TWT-capable station is associated with the mobile AP station in the always-on soft AP mode, operate the mobile AP station as a second station in an infrastructure mode associated with an infrastructure AP; and transmit, via the transceiver, infrastructure traffic to the infrastructure AP.
2. The mobile AP station of claim 1, wherein the one or more processors are further configured to enable the first TWT-capable station to request setting up the TWT period for the first TWT-capable station.
3. The mobile AP station of claim 2, wherein to enable the first TWT-capable station to request setting up the TWT period, the one or more processors are further configured to: set a TWT Required field to a predetermined value in a high efficiency operation element in a second beacon; in response to the TWT Required field being set to the predetermined value, receive a TWT setup request from the first TWT-capable station; in response to the TWT setup request, transmit a TWT schedule to the first TWT-capable station; wherein to inform the first TWT-capable station that the mobile AP station is not available outside of the TWT schedule, set a TWT Responder Power Save field in a TWT Element; and transmit the TWT element to the first TWT-capable station.
4. The mobile AP station of claim 2, wherein the one or more processors are further configured to remain active outside of the TWT period for the first TWT-capable station.
5. The mobile AP station of claim 1, wherein the one or more processors are further configured to dynamically adjust the TWT period for the first TWT-capable station based at least in part on a number of stations associated with the mobile AP station or a traffic load between the mobile AP station and one or more stations of the number of stations associated with the mobile AP station.
6. The mobile AP station of claim 1, wherein the one or more processors are further configured to dynamically adjust the wake window for unassociated stations based at least in part on a number of stations associated with the mobile AP station.
7. The mobile AP station of claim 1, wherein the one or more processors are further configured to dynamically adjust the TWT period for the first TWT-capable station based at least in part on a number of TWT-capable stations associated with the mobile AP station.
8. The mobile AP station of claim 1, wherein the one or more processors are further configured to dynamically adjust a duration of the infrastructure mode based at least in part on a traffic load of TWT-capable stations associated with the mobile AP station.
9. The mobile AP station of claim 1, wherein the one or more processors are further configured to dynamically adjust a duration of a next TWT during which a TWT-capable station unassociated with the mobile AP station associates with the mobile AP station based at least in part on a number

of TWT-capable stations associated with the mobile AP station.

10. A method for a mobile access point (AP) station, comprising: operating in an always-on soft AP mode, wherein the operating comprises implementing a target wake time (TWT) feature; transmitting a beacon that is followed by a wake window for stations unassociated with the mobile AP station, wherein the beacon comprises a broadcast TWT schedule that indicates a service period during which one or more TWT-capable stations unassociated with the mobile AP station associates with the mobile AP station; while operating the mobile AP station in the always-on soft AP mode enabling a first TWT-capable station to associate with the mobile AP station during the service period, wherein the mobile AP station provides soft AP functionality for the associated first TWT-capable station during a TWT period; while the first TWT-capable station is associated with the mobile AP station in the always-on soft AP mode, operating the mobile AP station as a second station in an infrastructure mode associated with an infrastructure AP; and transmitting infrastructure traffic to the infrastructure AP.

11. The method of claim 10, further comprising: enabling the first TWT-capable station to request setting up the TWT period for the first TWT-capable station.

12. The method of claim 10, further comprising dynamically adjusting the TWT period for the first TWT-capable station based at least in part on: a number of stations associated with the mobile AP station; or a traffic load between the mobile AP station and one or more stations of the number of stations associated with the mobile AP station.

13. The method of claim 10, further comprising dynamically adjusting the wake window for unassociated stations based at least in part on a number of stations associated with the mobile AP station.

14. The method of claim 10, further comprising dynamically adjusting a duration of the infrastructure mode based at least in part on a traffic load of TWT-capable stations associated with the mobile AP station.

15. A non-transitory computer-readable medium storing instructions that, when executed by a processor of a mobile access point (AP) station, cause the mobile AP station to perform operations, the operations comprising: operating the mobile AP station in an always-on soft AP mode comprising implementing a target wake time (TWT) feature; transmitting a beacon that is followed by a wake window for stations unassociated with the mobile AP station, wherein the beacon comprises a broadcast TWT schedule that indicates a service period during which one or more TWT-capable stations unassociated with the mobile AP station associates with the mobile AP station; while operating the mobile AP station in the always-on soft AP mode, enabling a first TWT-capable station to associate with the mobile AP station during the service period, wherein the mobile AP station provides soft AP functionality for the associated first TWT-capable station during a TWT period; while the first TWT-capable station is associated with the mobile AP station operating in the always-on soft AP mode, operating the mobile AP station as a second station in an infrastructure mode associated with an infrastructure AP; and transmitting infrastructure traffic to the infrastructure AP.

16. The non-transitory computer-readable medium of claim 15, wherein the operations further comprise: enabling the first TWT-capable station to request setting up the TWT period for the first TWT-capable station.

17. The non-transitory computer-readable medium of claim 16, wherein to enable the first TWT-capable station to request setting up the TWT period, the operations further comprise: setting a TWT Required field to a predetermined value in a high efficiency operation element in a second beacon; in response to the TWT Required field being set to the predetermined value, receiving a TWT setup request from the first TWT-capable station; and in response to the TWT setup request, transmitting a TWT schedule to the first TWT-capable station.

18. The non-transitory computer-readable medium of claim 15, wherein the operations further comprise dynamically adjusting the TWT period for the first TWT-capable station based at least in

part on: a number of stations associated with the mobile AP station; or a traffic load between the mobile AP station and one or more stations of the number of stations associated with the mobile AP station.

19. The non-transitory computer-readable medium of claim 15, wherein the operations further comprise dynamically adjusting the wake window for unassociated stations based at least in part on a number of stations associated with the mobile AP station.

20. The non-transitory computer-readable medium of claim 15, wherein the operations further comprise dynamically adjusting a duration of the infrastructure mode based at least in part on a traffic load of TWT-capable stations associated with the mobile AP station.
